

CHEMICAL ANALYSIS – PURITY & FORMULATIONS

KEY VOCABULARY

Pure substance: a single element or compound, not mixed.

Formulation: a useful mixture made to an exact recipe.

Melting point test: a pure substance melts at one sharp temperature.

Impurity: lowers melting point and widens the range.

COMMON MISCONCEPTIONS

~~— 'Pure' means safe or natural.~~

✓ In chemistry pure means a single substance.

~~— Impurities raise the melting point.~~

✓ Impurities lower it and spread it over a range.

~~— A formulation is a pure substance.~~

✓ A formulation is a carefully made mixture.

PURE VS IMPURE

Pure water melts sharply at 0°C.

Pure substances have a fixed melting point.

An impurity lowers and broadens the melting point.

Formulations: paints, fuels, medicines, alloys.

Each component has a measured amount and purpose.

1

DEFINE PURE

State what a chemist means by a **pure substance**. [1 mark]

2

TEST FOR PURITY

Explain how a melting point test shows whether a substance is pure. [2 marks]

3

FORMULATION

State what is meant by a **formulation** and give an example. [2 marks]

4

EFFECT OF IMPURITY

Describe how an impurity changes the melting point. [2 marks]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

A pure substance melts at a ____ temperature; an impurity melts over a ____.

sharp

range

lower

higher

6

EVERYDAY EXAMPLES

Give two everyday formulations and their purpose. [2 marks]